

Judge Q&A

Eco-Loop / TwinPilot · HirePro Honeywell · <https://github.com/alphasafal/Twincity-/tree/ecoloop-hackathon-final>

Judge Q&A – prepared answers

Did the LLM produce the 4.98% savings?

The primary quantitative result comes from the reproducible **Path A** deterministic optimisation experiment (comfort-zero policy). The LLM operates as a **supervisory agent** through MCP (Path C), selecting contextual strategies and proposing bounded actions. Numeric setpoints are computed by the deterministic optimiser; SafetyShield retains final authority. Path C hybrid runs prove the LLM is materially involved in closed-loop decisions without replacing Path A as the authoritative claim.

Why not use only rules?

Rules are responsible for safety, but they do not provide flexible contextual planning, explanation or tool-based adaptation. The LLM selects strategies using occupancy, weather, thermal state and recent outcomes; deterministic components execute and enforce them safely.

Why is total saving only 1.31%?

HVAC represents only part of this model’s total consumption. Lighting and equipment loads were unchanged. The controlled HVAC portion fell by **4.98%**, producing a **1.31%** total-building reduction.

How do you prove the action reached EnergyPlus?

Runtime traces record observation → proposal → SafetyShield decision → executed `Clg-SetP-Sch` value → following EnergyPlus state for consecutive timesteps (`final-release/evidence/final-actuator-trace.csv`, `final-next-state-proof.md`, hybrid `stage\_log.jsonl`).

Is this connected to a real building?

No. The prototype closes the loop against an EnergyPlus digital building. Physical BMS integration is the next deployment step.

What happens when AI fails?

The system detects Ollama or MCP failure and activates deterministic fallback. AI availability is not required for safe continued operation.

What are Paths A / B / C?

| Path | Role |

|-----|-----|

| A | Authoritative comfort-zero savings (deterministic optimiser) |

| B | LLM structured setpoint proposals via separate MCP stdio |

| C | Hybrid: LLM selects ECM/strategy → optimiser → SafetyShield → actuator → self-correction |